University of Nottingham
UK | CHINA | MALAYSIA

COMP-2032

Introduction to Image Processing

Lecture 1A
Introduction

Learning Outcomes

WHAT

- Image Processing
- Module Content

IDENTIFY

Digital Images

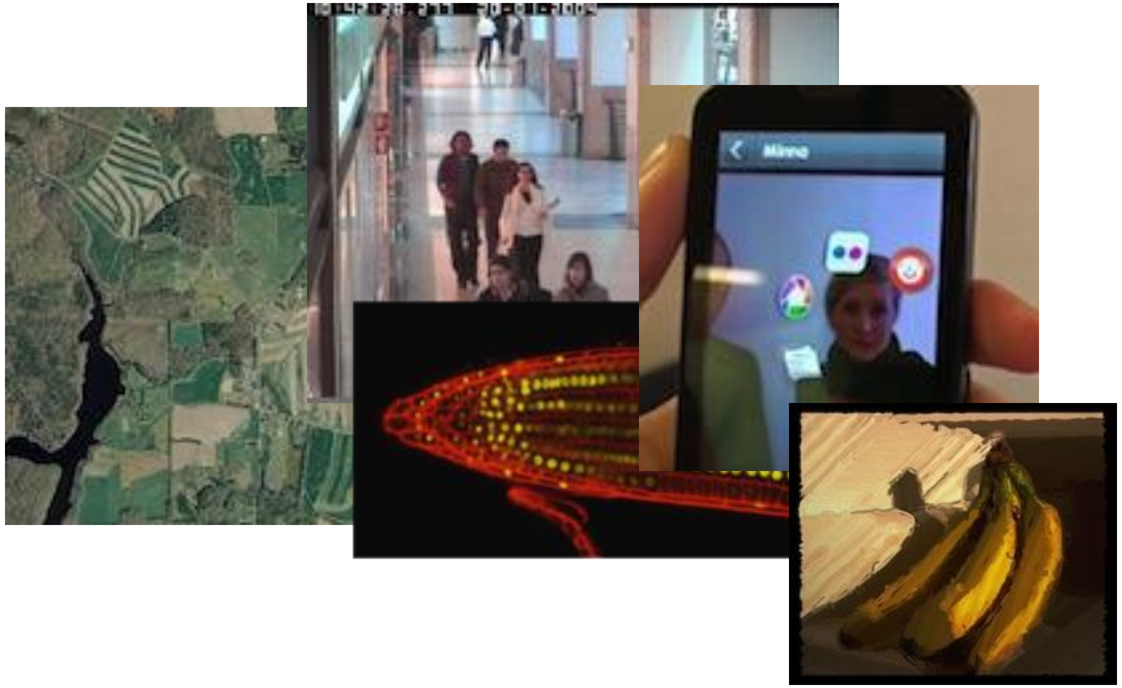
What is Image Processing?

Let’s Debunk!

Image Processing

Collection of topics & techniques related to the use of computers to perform the following actions on digital images

- *Acquire*
- *Store*
- *Manipulate*
- *Model*
- *Analyse/Interpret*
- *Display*



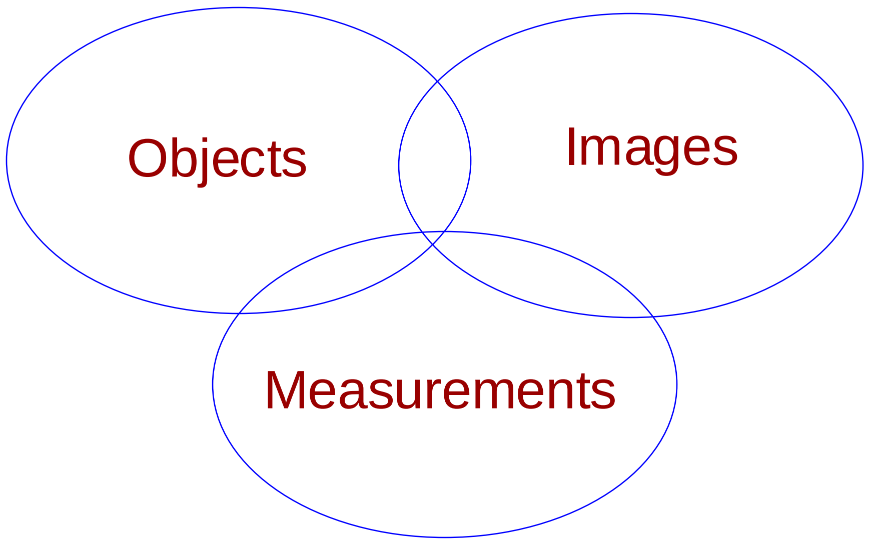
Generic techniques that are applied to images from most sources

ACK: Prof. Tony Pridmore, UNUK

Know Your Limitations

4 Terms are often used together, and some are sometimes confused

1. *Image Processing*
2. *Image Analysis*
3. *Computer Vision*
4. *Computer Graphics*



- All share representations, underlying mathematics & some algorithms
- Their goals are very different

This is an Image Processing module with a little Image Analysis

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Image Analysis

What is the difference?

- Concerned with making quantitative measurements on images:
  - Image acquisition is constrained so that image measurements are a proxy for some real-world value
  - Sits between image processing and computer vision

Objects → Images → Measurements

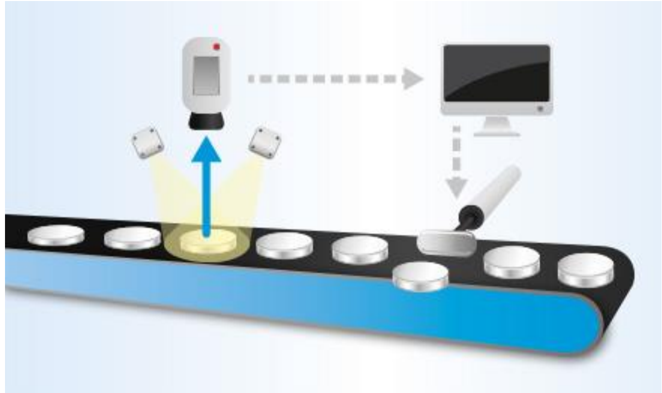

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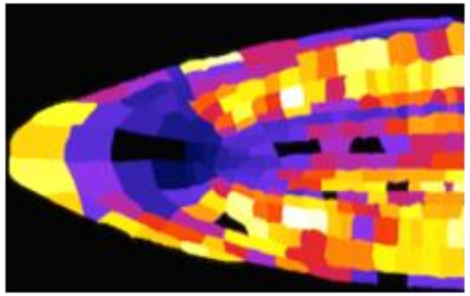
Image Analysis

- Solutions are application specific
- Uses generic operations where possible, but takes an engineering more than scientific approach
- Many application areas:

- Medical
- Scientific
- Industrial
  - Food, textiles, manufacturing….



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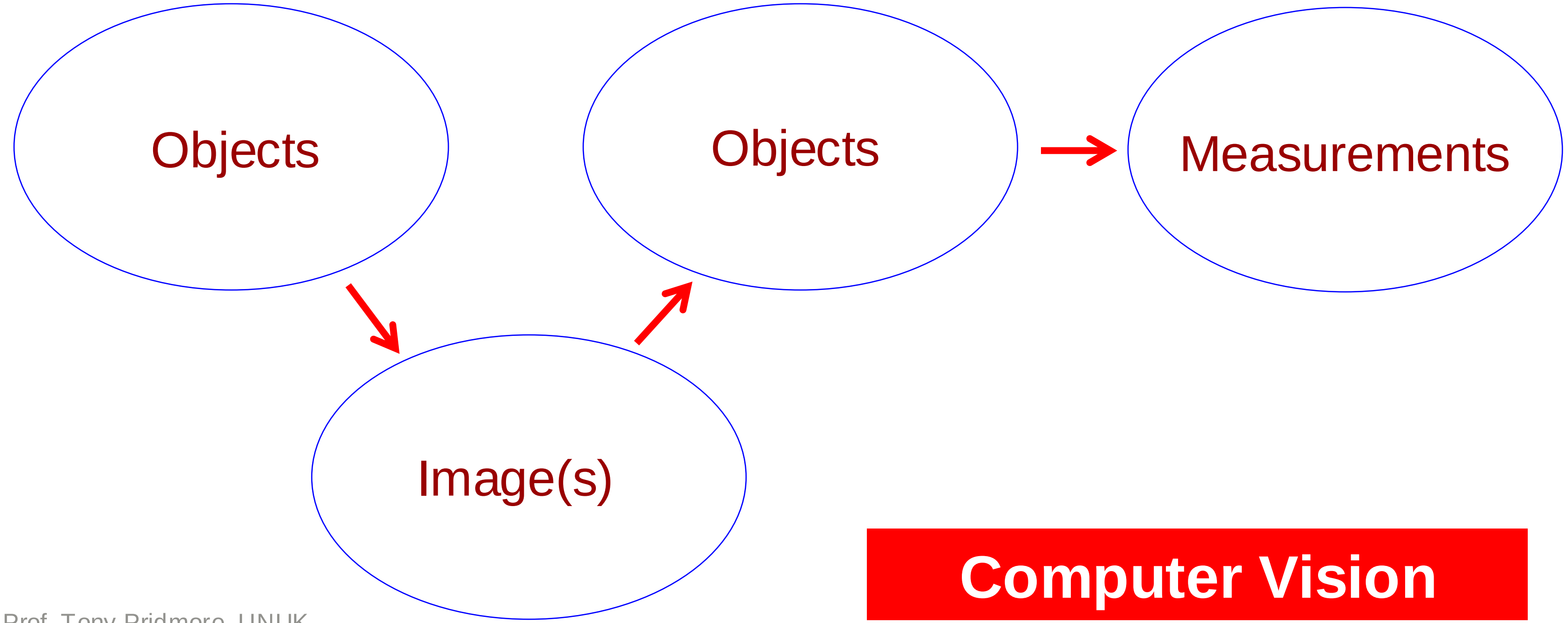
It's NOT Computer Vision

Aims to invert image formation & recover information about the viewed world: 3D shape, motion, identity…

What is the difference?



Computer Vision

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It's NOT Computer Graphics

Focus is on creating images from object models

What is the difference?

- Lighting and shading modelling
- Volume modelling
- Curve and surface modelling
- Visibility modelling
- Texture synthesis
- Character animation
- Modelling terrain, liquids, fire/smoke, cloth, hair/fur, feature, skin etc.

Computer Graphics

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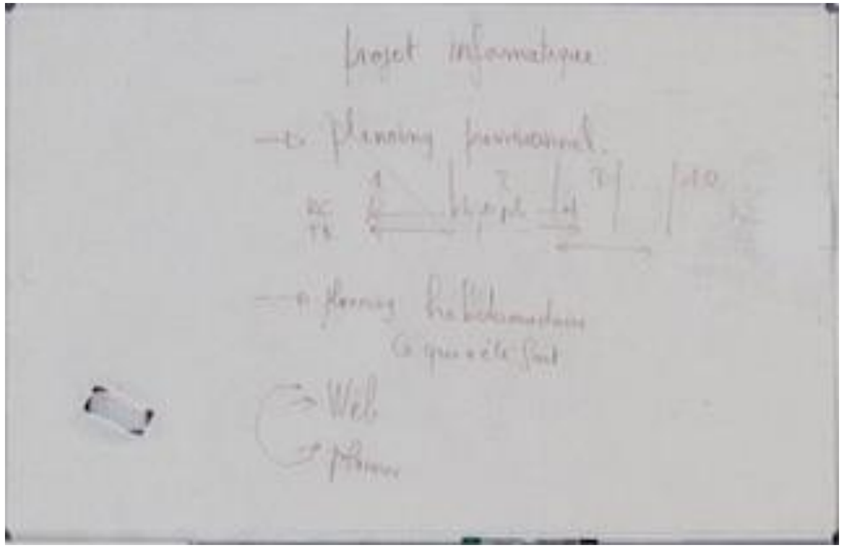
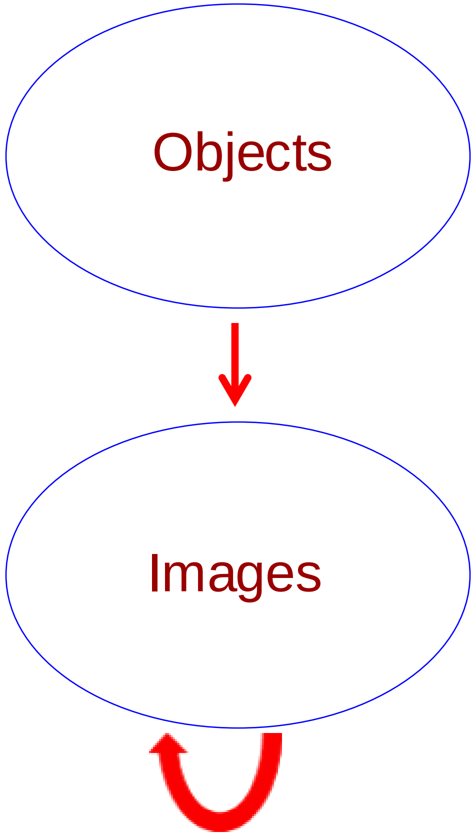
Image Processing

REMEMBER

- Image(s) in, image(s) out
- Key information more easily seen/extracted
- More aesthetically pleasing





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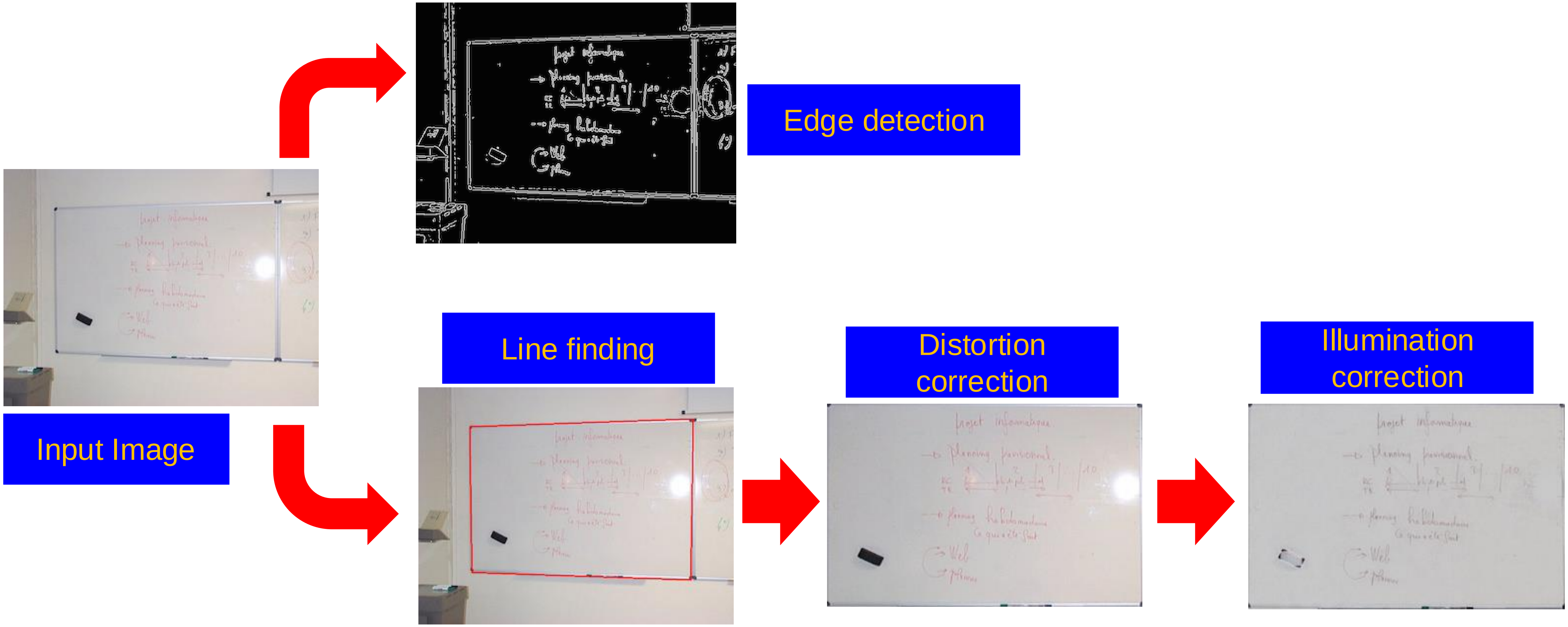
It’s a toolkit



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Primary Aims

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Introduce the fundamentals of digital image processing – theory & practice

Gain practical experience in writing programs to manipulate digital images

Lay the foundation for studying advanced topics in related fields

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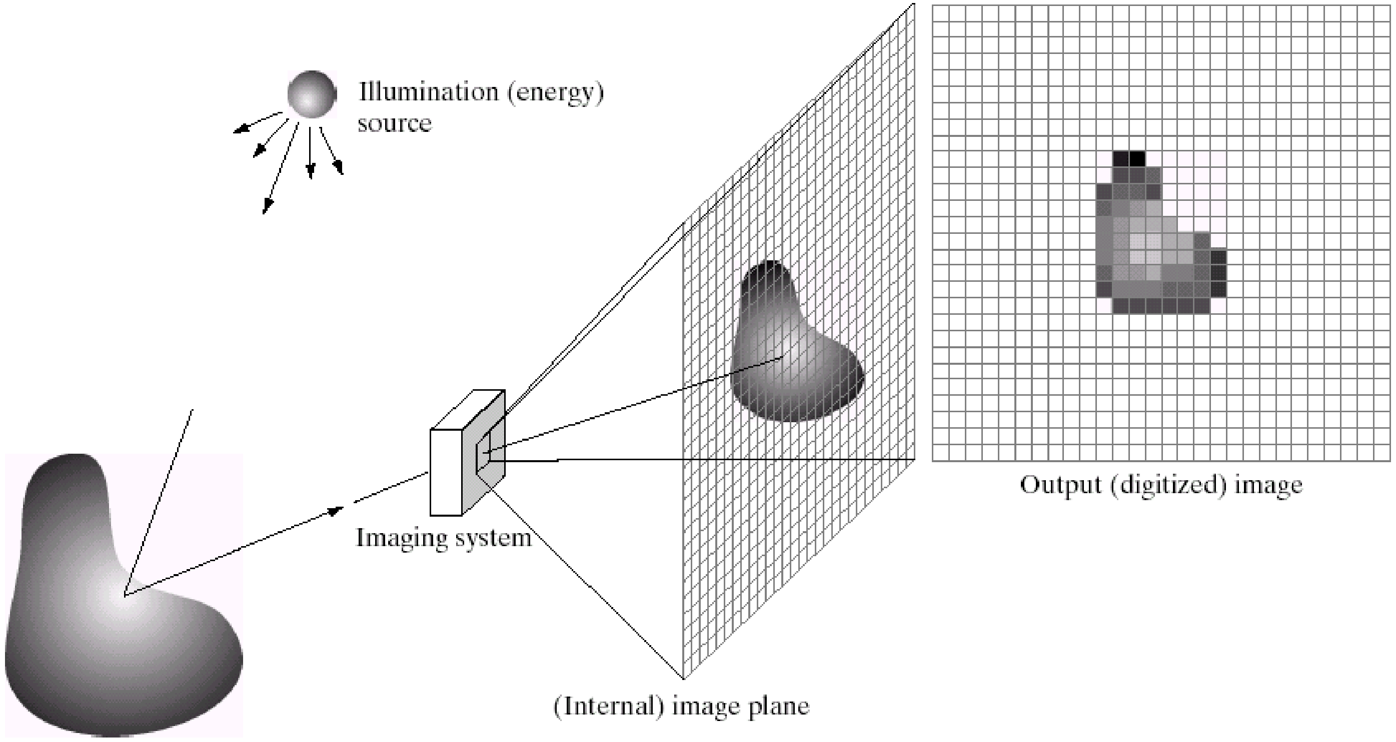
Module Content

More SECRETs Revealed!

Image Formation | **Acquisition** | **Colour Representation**



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Digital Images

A common, low-level representation of the viewed world

123 33 234 45
67 90 12 134 34
56 89 54 67 98
111 56 67 90 65
34 ….

Pixel values represent the brightness and colour of the viewed objects, but given no indication of what object, e.g., books, monitors, these numbers refer to – hence low-level

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Redundancy & image compression – efficiently represent image data for storage (*minimise disk space*) & communication (*minimise network bandwidth*)

245,760 bytes

69,632 bytes

5,951 bytes

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Image manipulation – noise removal, smoothing, sharpening, contrast enhancement, changing the appearance of an image, etc.

Noise reduction

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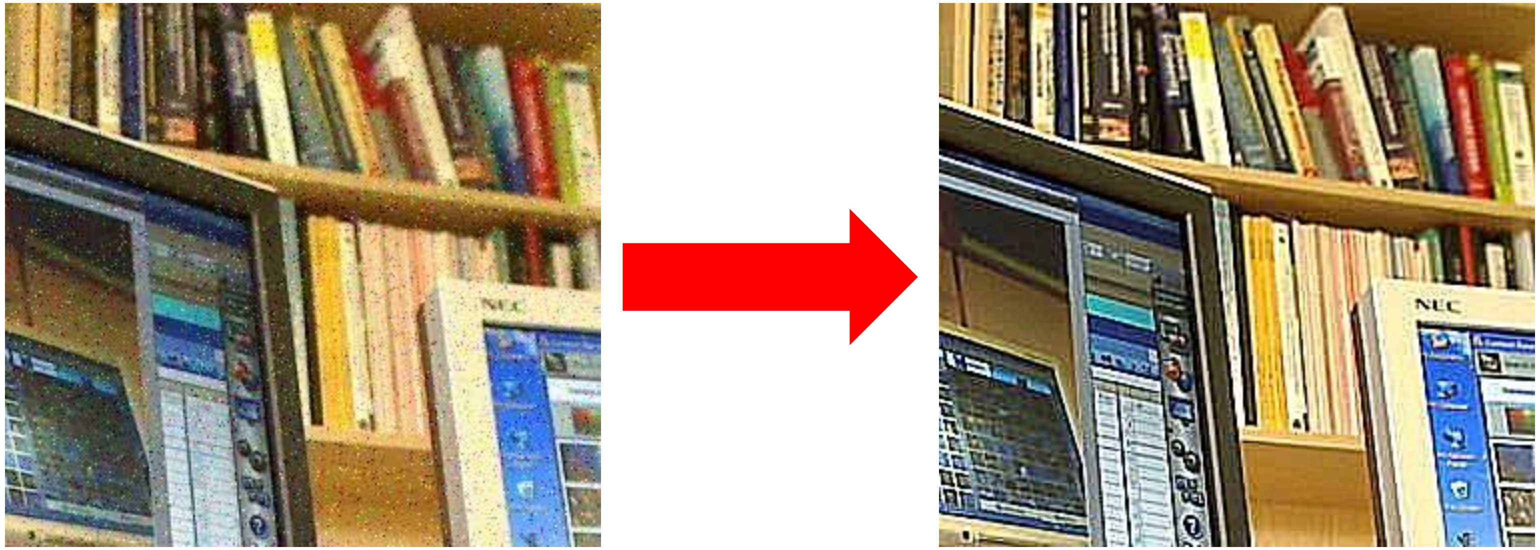
Image manipulation – noise removal, smoothing, sharpening, contrast enhancement, changing the appearance of an image, etc.



Sharpening

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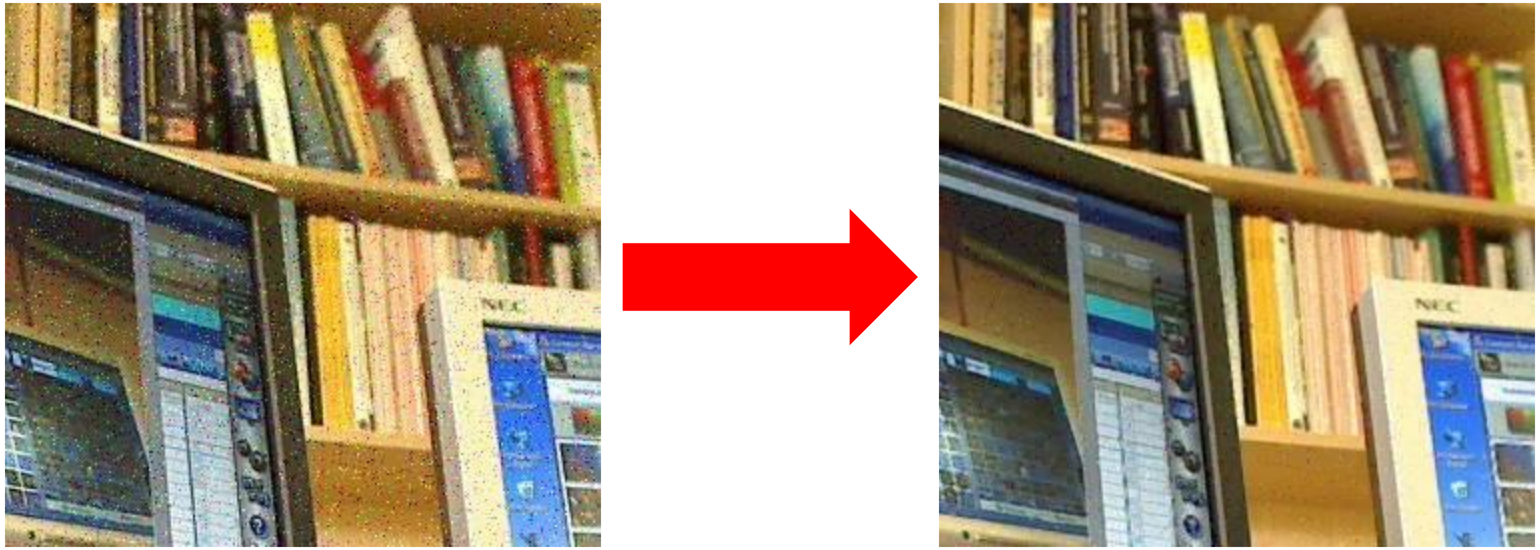
Image manipulation – noise removal, smoothing, sharpening, contrast enhancement, changing the appearance of an image, etc.



Smoothing

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Image manipulation – noise removal, smoothing, sharpening, contrast enhancement, changing the appearance of an image, etc.

Contrast Enhancement

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- **Focus** on spatial domain methods (*operating directly on the image*)
    - Point operations
    - Area operations

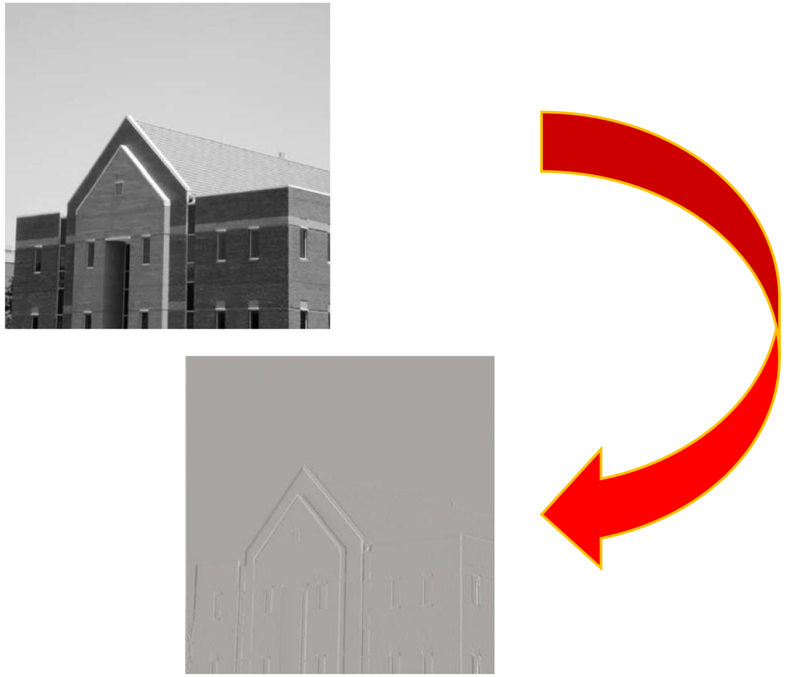

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Image manipulation – noise removal, smoothing, sharpening, contrast enhancement, changing the appearance of an image, etc.

Changing image appearance

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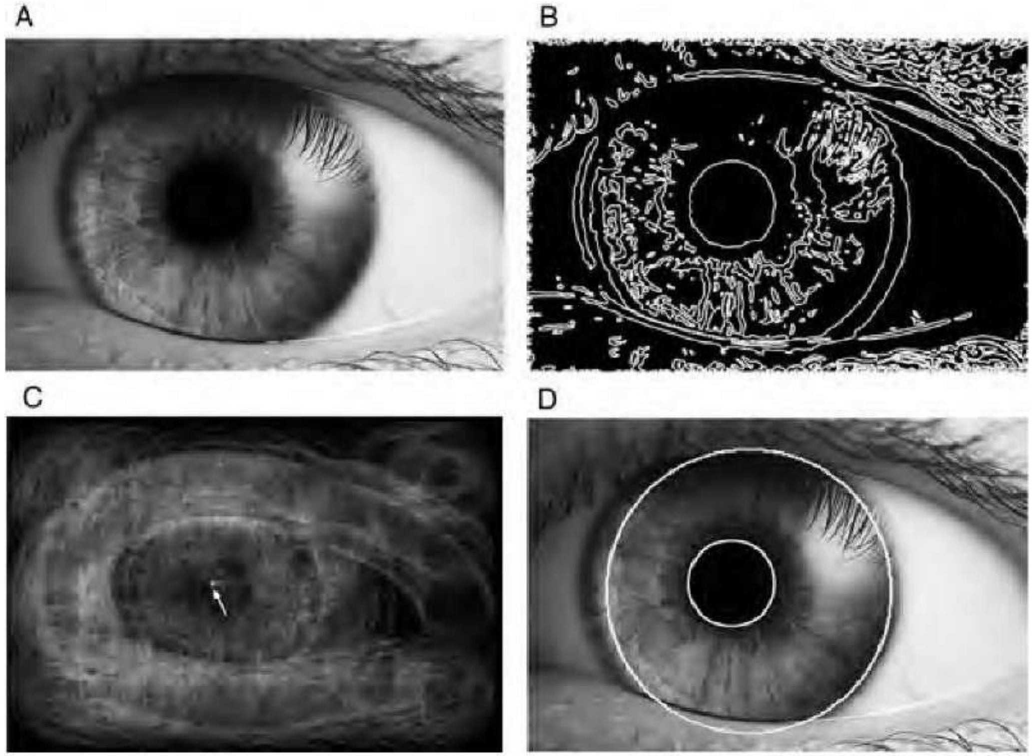
Finding geometric objects



The Hough Transform

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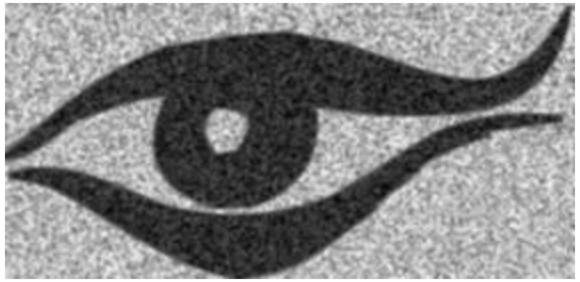
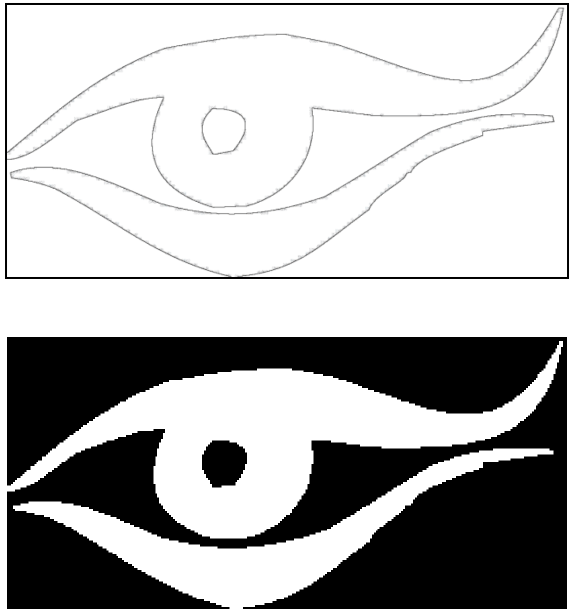
Edge detection **Image segmentation**





- Underlying theory
- Some useful algorithms

A step towards image analysis & computer vision

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Interactive Segmentation - Superpixels

- An increasingly popular intermediate representation
- Fewer data points = less work

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Geometric operations – manipulate the array structure underlying image, not just the pixel values

Rotation, translation, scaling and affine transformations

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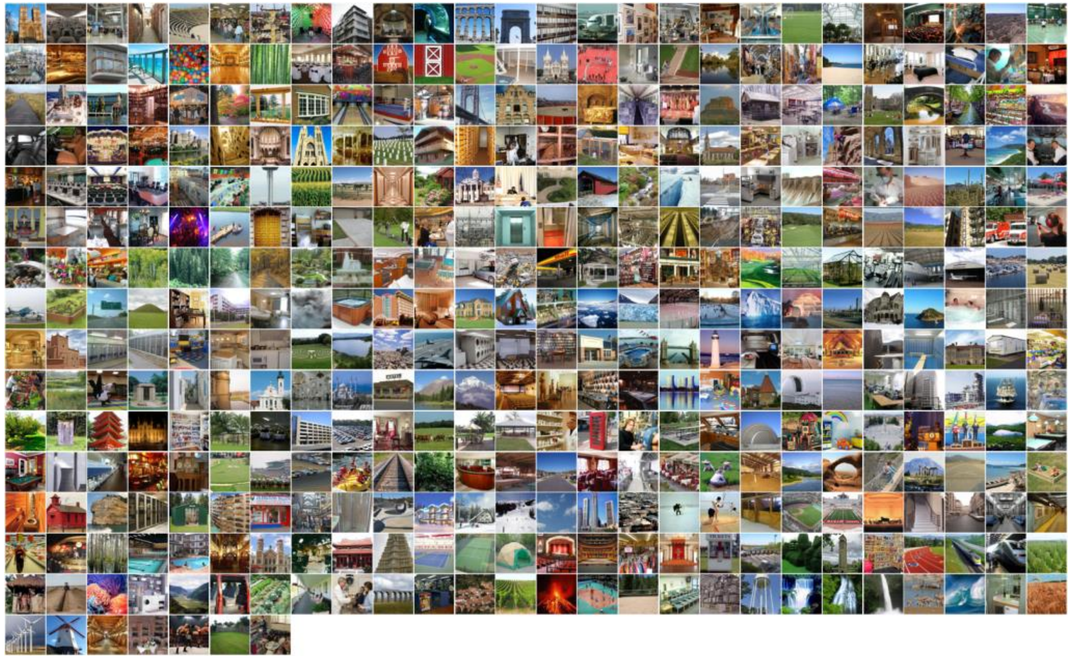
Some Applications

Content-based Image Retrieval

Show me all the image like this:

In here:



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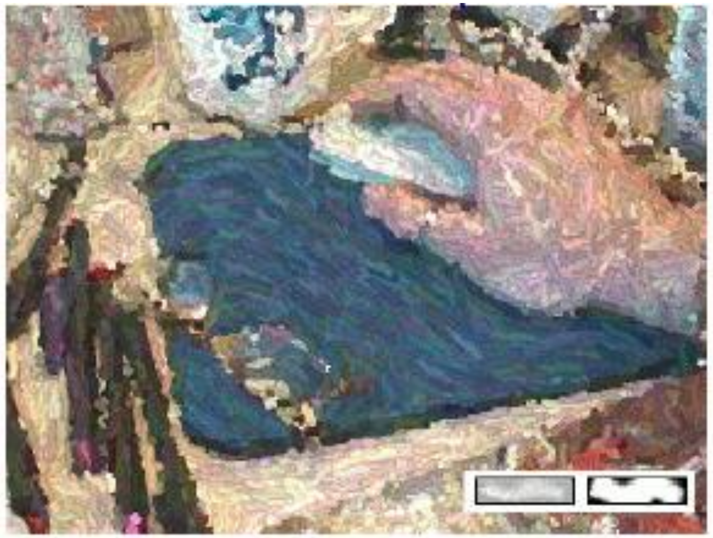
Some Applications

Painterly Rendering

- Process images to give a painted feel
- Aims to reproduce a particular artist or movement's style, e.g., **Impressionism**



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Some Applications

Interactive Tools & Compositing

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Assessment

Very IMPORTANT

COMP2032 – Assessment (100% CW)

- Programming + **2000-word** conference paper
  - Python application
  - Literature review, explanation & evaluation of results

Group

- In lab test
  - Answer **ALL** questions

Individual

Summary

1

- Image Processing
- Module Content

2

Digital Images

Questions

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NEXT:

Digital Images and Point Processes